

House Price Prediction

A Regression-Based Data Science Project

IBM DATA SCIENCE CAPSTONE PROJECT

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Domain	Real Estate / Property Valuation
Tech Stack	Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Dataset	1,800 residential properties across 6 neighborhoods

This report predicts residential property prices from structural, locational, and quality features using regression modeling, and translates the results into pricing insights for buyers, sellers, and agents.

1. Business Problem

Accurately pricing a residential property is critical for sellers, buyers, and real-estate platforms alike — overpricing stalls a sale, underpricing loses value. The objective of this project is to build a regression model that predicts a house's market price from its physical characteristics (area, bedrooms, bathrooms), location factors (neighborhood, distance to city center, school rating, crime index), and quality indicators (pool, garden, renovation status, age).

The dataset covers **1,800 properties** with a median price of **\$272,250** and an average price of **\$275,556**.

2. Data Collection & Preparation

Property records include structural attributes (area_sqft, bedrooms, bathrooms, stories, garage_spaces), location attributes (neighborhood, distance_to_center_km, school_rating, crime_index), and condition attributes (age_years, has_pool, has_garden, renovated). The data was checked for missing values, duplicate records, and outliers — the cleaned dataset had **0 missing values** and **0 duplicate rows**.

3. Exploratory Data Analysis

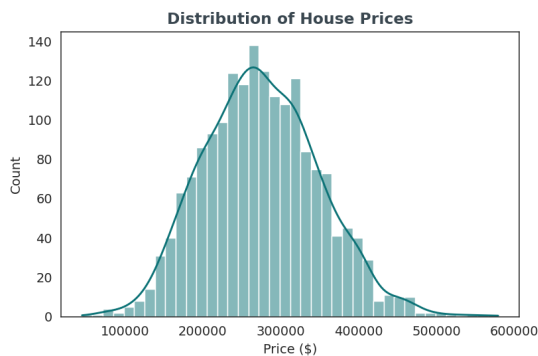


Fig 1. Price distribution is right-skewed, typical of housing markets.



Fig 2. Living area shows a strong positive relationship with price.

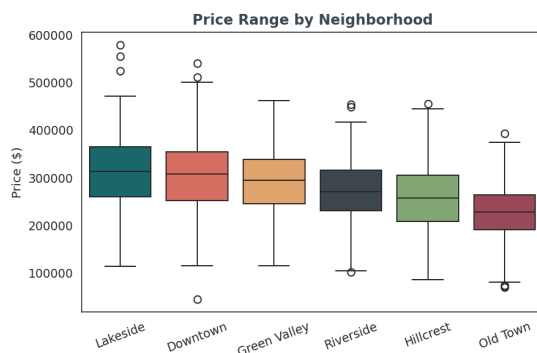


Fig 3. Lakeside and Downtown command the highest median prices.

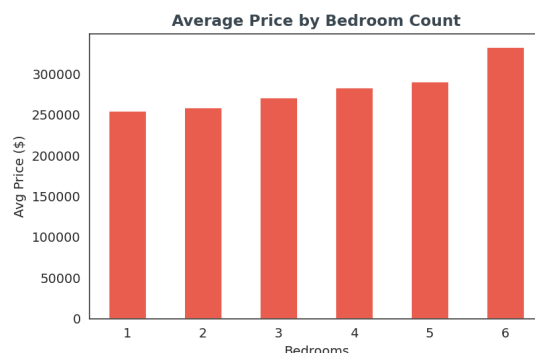


Fig 4. Average price rises steadily with bedroom count.

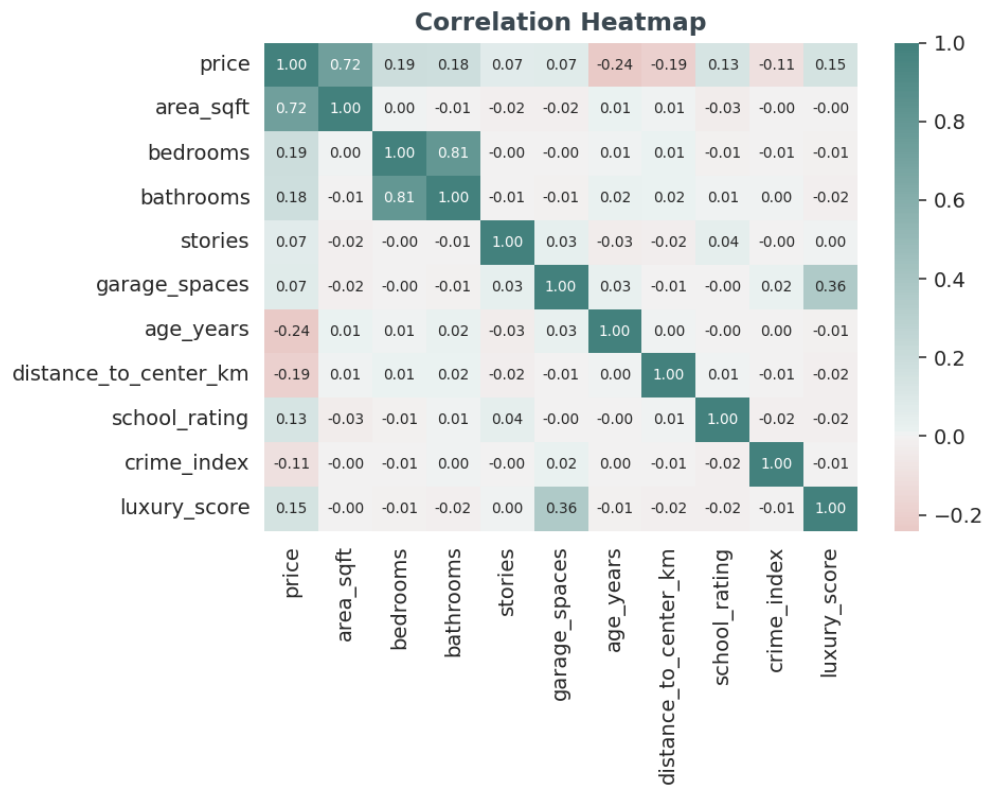


Fig 5. Correlation heatmap across numeric property features.

Key EDA Findings

- Living area (sqft) is the single strongest driver of price, showing a near-linear relationship.
- Neighborhood carries a significant price premium — Lakeside and Downtown properties sell well above Old Town for comparable specs.
- Price declines with distance from the city center and with crime index, and rises with school rating.
- Older, unrenovated homes sell at a discount relative to newer or recently renovated ones.

4. Feature Engineering

- **total_rooms** — bedrooms + bathrooms, a simple size proxy.
- **luxury_score** — composite of pool, garden, renovation, and 2+ car garage, capturing overall property quality.
- **age_category** — age_years bucketed into New / Recent / Established / Old bands.
- **price_per_sqft** — computed for reporting only and excluded from modeling to avoid target leakage.

Neighborhood and age_category were one-hot encoded; numeric features were standardized for the linear models. The final feature matrix contained **22 features**.

5. Model Training & Evaluation

The data was split into train and test sets (1440 / 360, 80/20). Four regression models were trained and compared using RMSE, MAE, R2 Score, and MAPE:

Model	RMSE	MAE	R2 Score	MAPE (%)
Linear Regression	\$19,990	\$15,843	0.925	6.52%
Ridge Regression	\$19,993	\$15,846	0.925	6.51%
Gradient Boosting	\$23,464	\$17,989	0.896	7.08%
Random Forest	\$31,310	\$24,831	0.815	9.67%

Linear Regression performed best, explaining **92.5%** of the variance in price (R2) with a mean absolute error of about **\$15,843** and a typical error of **6.5%** of the true price. The strong performance of the linear model suggests price in this market is largely explained by a linear combination of the given features, with limited nonlinear interaction effects.

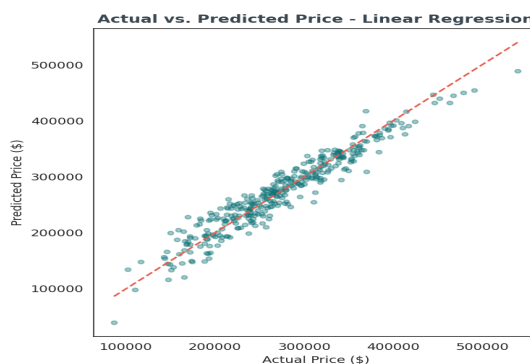


Fig 6. Predicted vs. actual price — points cluster tightly along the diagonal.

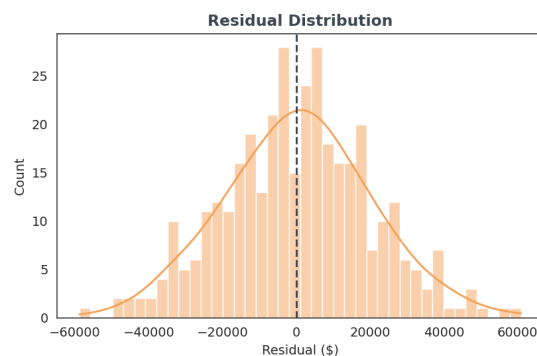


Fig 7. Residuals are roughly centered at zero with no strong skew.

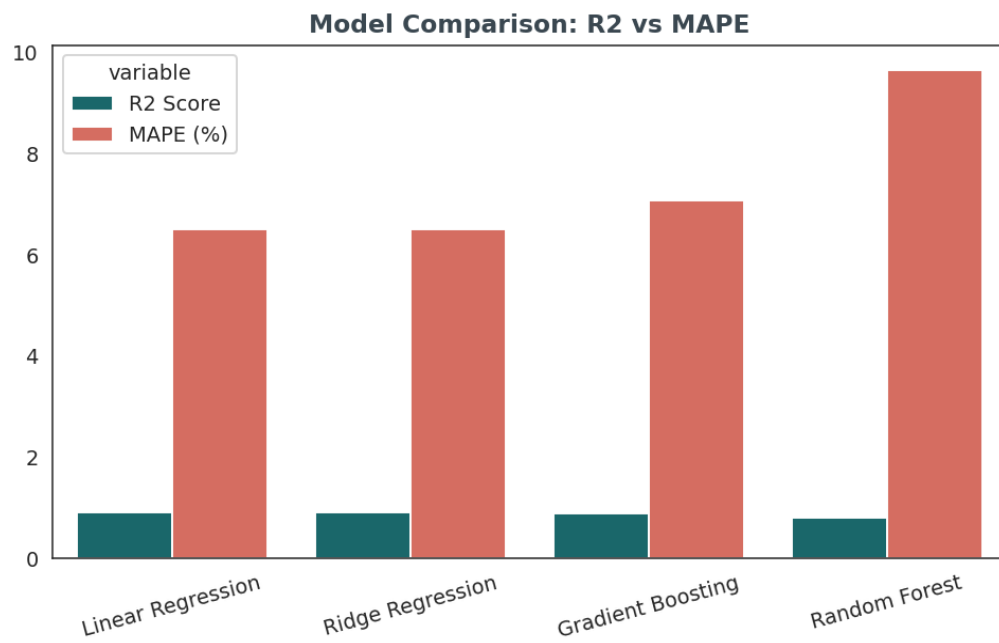


Fig 8. R2 and MAPE across all four candidate models.

6. Feature Importance

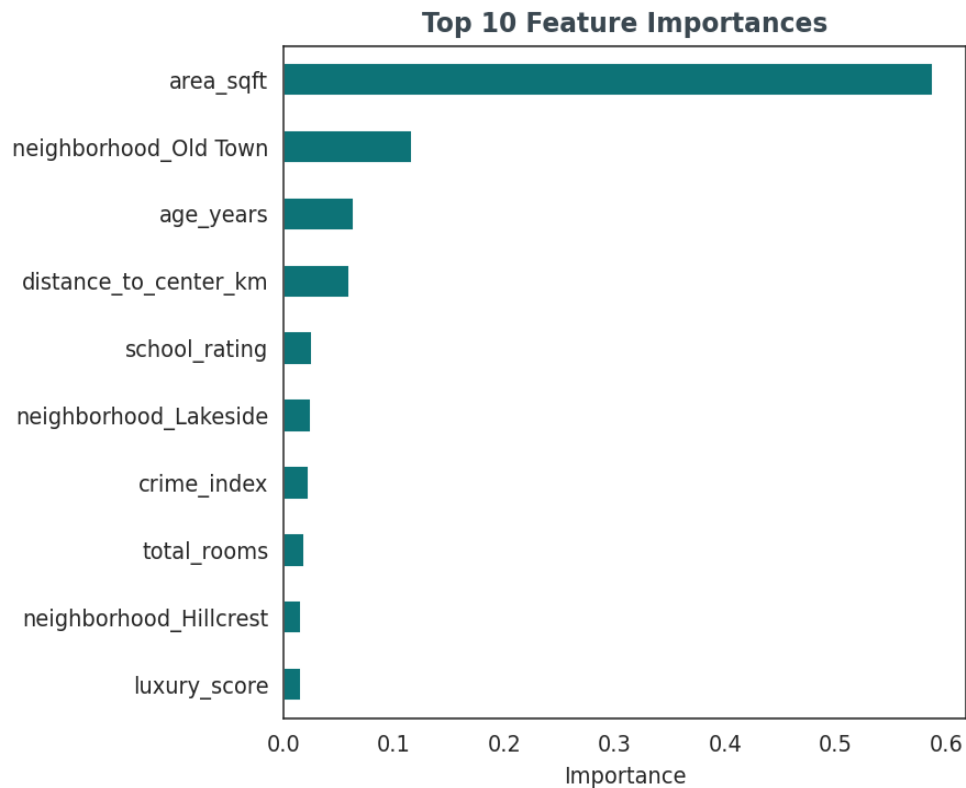


Fig 9. Top 10 features driving predicted price.

The dominant price drivers were **area_sqft**, **neighborhood_Old Town**, **age_years**, **distance_to_center_km**, **school_rating** — confirming that size, location premium/discount, age, and neighborhood safety/school quality jointly determine market value.

7. Business Recommendations

- **Use the model as a pricing guide, not a final price.** With ~6.5% typical error, list prices should still allow room for negotiation and local agent judgment.
- **Highlight neighborhood value in listings.** Since neighborhood is a top price driver, marketing should foreground school ratings and safety statistics for premium areas.
- **Advise renovation ROI.** Renovated status and luxury features (pool, garden, garage) meaningfully lift price — sellers of older homes may benefit from targeted upgrades before listing.
- **Flag under/over-priced listings.** Comparing the model's predicted price to the asking price can surface listings that are mispriced relative to the market.

8. Possible Extensions

This project can be extended with geospatial features (actual latitude/longitude and nearby amenity density), a Streamlit price-estimator app, and hyperparameter tuning (GridSearchCV) on the ensemble models for further

error reduction.

Dataset: synthetically generated to reflect realistic residential pricing dynamics for demonstration purposes.